

Prasanna B

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EDUCATION

Amrita Vishwa Vidyapeetham

Coimbatore, Tamil Nadu

B.Tech in Computer Science and Engineering — CGPA: 8.70 | 4th Sem SGPA: 8.42

Aug 2024 – May 2028

- Relevant Coursework: Data Structures, Algorithms, Operating Systems, DBMS, User Interface Design

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, Haskell

Backend: FastAPI, REST APIs, SQLAlchemy

Frontend: React, HTML5, CSS3

Databases: SQLite, Neo4j

Tools: Git, Linux, Bash, Docker

Core Concepts: DSA, Graph Algorithms, Probabilistic Data Structures, System Design

PROJECTS

dupegun | *Python, SHA-256, Bloom Filters, TUI, REST*

2026 🔄

- Built a cross-platform CLI tool for duplicate file detection using a 3-pass hashing engine (size grouping, partial hashing, full SHA-256).
- Implemented interactive TUI, browser-based local web UI, real-time folder monitoring, and perceptual image duplicate detection.
- Designed plugin system for custom deduplication strategies; supports dry-run mode, scheduled automation, and export to JSON/CSV/HTML.
- Published as a PyPI package with optional feature sets: `tui`, `watch`, `serve`, `images`, `schedule`.

Eco-Fix | *React, Firebase, JavaScript, HTML, CSS*

2025 🔄

- Developed a responsive web application focused on promoting eco-friendly waste management and reporting solutions.
- Implemented Firebase Authentication and Firestore Database for secure user management and real-time data handling.
- Designed interactive user interfaces with React and modern frontend technologies to enhance user experience and accessibility.

FleetTrack | *ESP32, React, Node.js*

2026 🔄

- Developed a full-stack real-time fleet tracking system integrating ESP32-based GPS hardware with a React live dashboard.
- Implemented real-time vehicle location updates, route visualization, and backend telemetry processing.
- Designed scalable architecture for continuous monitoring and fleet data management.

G-Wallet | *FastAPI, PostgreSQL*

2026 🔄

- Engineered a secure wallet system with ACID-compliant transaction handling and concurrency-safe batch processing.
- Implemented authentication, audit logging, and admin-based verification workflow to reduce fraudulent activities.
- Designed modular REST APIs with robust backend validation mechanisms.

Bloom Filter URL Filtering System | *Python, HTML*

2025 🔄

- Implemented a custom Bloom Filter to evaluate probabilistic URL filtering performance.
- Analyzed memory usage versus false-positive trade-offs for scalable filtering systems.
- Demonstrated efficiency of probabilistic data structures in high-volume lookup environments.

Neo4j Academic Graph Network | *Python, Neo4j*

2025 🔄

- Designed graph-based schema to model students, courses, and relationships using Neo4j.
- Implemented efficient CRUD operations and CSV data ingestion for academic datasets.
- Optimized Cypher queries for faster traversal and relationship analysis.

Mines | *Java, C++, Graph Algorithms*

2026 🔄

- Developed an algorithm-driven Minesweeper engine focusing on deterministic solving strategies and constraint propagation.
- Implemented graph traversal and logical inference techniques to optimize mine detection accuracy.
- Designed modular architecture enabling extensible board sizes and AI-based solution validation.

PUBLICATIONS

Towards Sustainable Charcoal Production from Prosopis Juliflora

Integrating Local Knowledge and Clean Technology

2025

- Study focused on charcoal production in Adancheri village, Tamil Nadu, highlighting environmental, health, and socioeconomic challenges of traditional earth mound kilns.
- Applied Participatory Rural Appraisal (PRA) tools including resource mapping, seasonal calendars, and stakeholder consultations to assess limitations.
- Proposed improved kiln technologies, capacity-building initiatives, and sustainable resource management strategies for local communities.

LEADERSHIP & RECOGNITIONS

NSS Unit-3 Student Coordinator

Amrita Vishwa Vidyapeetham

- Led community outreach initiatives and sustainability programs.

Aug 2024 – Present

Coimbatore, Tamil Nadu

Active Member

Youth United Council of India

- Participated in education and environmental volunteer activities; technical member.

Aug 2024 – Present

Coimbatore, Tamil Nadu

Hackathons

2024 – 2026

- Participated in Build2Break and Adobe India Hackathon.